Benjamin (Ben) Rachunok

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Academic Appointments

Assistant Professor — **Industrial and Systems Engineering**
North Carolina State University, Raleigh, NC — *2023–*

Graduate Faculty — **Operations Research**
North Carolina State University, Raleigh, NC — *2023–*

R&D Intern — **Center for Computing Research**
Sandia National Labs, Albuquerque, NM — *2017–2020*
Supervisor: Dr. Andrea Staid

Education & Training

Postdoctoral Researcher — **Civil and Environmental Engineering**
Stanford University, Stanford, CA — *2020-2022*
Advisor: Dr. Sarah Fletcher

Ph.D. — **Industrial Engineering**
Purdue University, West Lafayette, IN — *2016-2020*
Advisor: Dr. Roshanak Nateghi

Bachelor of Science — **Industrial and Systems Engineering**
North Carolina State University, Raleigh, NC — *2011–2016*

Research Support

Collaborations in Artificial Intelligence and Geosciences (CAIG) — **National Science Foundation**
$389,206 NC State / $884,606 total, Co-PI — *August 2026-July 2029*
Artificial Intelligence-Driven Optimization of Coastal Morphodynamics and Infrastructure Resilience
Collaborative PIs: Katherine Anarde (NC State), Brad Murray (Duke), Laura Moore (UNC-Chapel Hill)

Infrastructure Systems and People — **National Science Foundation**
$80,054 NC State / $697,903 total, PI — *September 2026-August 2029*
AI-Enabled Decision Support for Rate Setting and Infrastructure Investment in Decentralized Water Utilities under Drought
Collaborative PIs: Marta Zaniolo (Duke), Eric Edwards (UC Davis), Mehdi Nemati (UC Riverside)

North Carolina Collaboratory — **State of North Carolina**
$99,820, Lead PI — *May 2025-December 2026*
North Carolina Data Center Water Use Analysis

North Carolina Collaboratory — **State of North Carolina**
$99,719, Lead PI — *May 2025-December 2026*
North Carolina Power System Restoration Optimization

Digitizing Utilities Prize, Phase 1 — **Department of Energy**
$75,000, Lead PI — *June 2024-June 2025*
North Carolina Short-term Power Outage Prediction

R.L. Rabb Science and Society Fund — **North Carolina State University**
$7625, Lead PI — *June 2024-June 2025*
North Carolina Climate & Health Systems Impacts Symposium

Research Collaboration Fund — **University Global Partnership Network**
$30,000, Lead NC State PI — *August 2024-August 2025*
Leveraging AI for Real-time Prediction of Extreme Weather Events to Foster Climate Resilience: a Global Research Alliance

UPS Endowment Fund — **Stanford University**
$100,000, Co-PI — *September 2022-August 2023*
Assessing Drought Impacts on Urban Water Affordability and Access

Peer-Reviewed Journal Articles

* indicates equal contributions, Underline indicates an advisee

[30] Y. Essus, **B. Rachunok**, T. Logan, K. Hardaway, and D. Bristow. "Towards Operationalizing Ecological Resilience for Infrastructure Decision-Making". In: *Risk Analysis* (2026). DOI: 10.1111/risa.70316.

[29] A. Helmrich, J. R. Wolfinbarger, K. Hardaway, **B. Rachunok**, and M. Haraguchi. "Flexible infrastructure design for changing conditions". In: *Environmental Research: Infrastructure and Sustainability* (2026). DOI: 10.1088/2634-4505/ae9c6f.

[28] P. Leavitt, F. Livingston, B. McConnell, and **B. Rachunok**. "Anticipating Household Rescue Demand in Hurricanes Using Socio-Demographic Data and Machine Learning". In: *International Journal of Disaster Risk Reduction* 145 (2026), p. 106405. DOI: 10.1016/j.ijdrr.2026.106405.

[27] T. Logan, **B. Rachunok**, K. Hardaway, D. Bristow, A. Reilly, D. Johnson, J. Johansson, G. Cremen, J. Boakye, and P. Schweizer. "Advancing systemic risk assessment for complex, interdependent systems: a research agenda". In: *Risk Analysis* 46.5 (2026). DOI: 10.1111/risa.70243.

[26] Y. Essus and **B. Rachunok**. "A Computational Framework for Modeling Access to Services and the Resilience Impacts of Electric Vehicles During Blackouts". In: *IEEE Access* (2025). DOI: 10.1109/ACCESS.2025.3629567.

[25] K. Hardaway, K. Best, and **B. Rachunok**. "Leveraging the American Housing Survey to Quantify Risk and Resilience". In: *Risk Analysis* (2025). DOI: 10.1111/risa.70100.

[24] Y. Essus and **B. Rachunok**. "Electric Vehicles Limit Equitable Access to Essential Services During Blackouts". In: *NPJ Urban Mobility* (2024). DOI: 10.1038/s44333-024-00007-3.

[23] **B. Rachunok**, A. Verma, and S. Fletcher. "Predicting and understanding residential water use with interpretable machine learning". In: *Environmental Research Letters* 19.1 (2024). DOI: 10.1088/1748-9326/ad1434.

[22] J. Skerer, A. Verma, M. Edwards, **B. Rachunok**, and S. Fletcher. "Alternative household water affordability metrics using water bill delinquency behavior". In: *Environmental Research Letters* (2024). DOI: 10.1088/1748-9326/ad5609.

[21] A. Nayak, **B. Rachunok**, B. Thompson, and S. Fletcher. "Socio-hydrological impacts of rate design on water affordability during drought". In: *Environmental Research Letters* 18.12 (2023). DOI: 10.1088/1748-9326/ad0994.

[20] **B. Rachunok** and S. Fletcher. "Socio-hydrological drought impacts on urban water affordability". In: *Nature Water* 1.1 (2023). DOI: 10.1038/s44221-022-00009-w.

[19] J. Bennett, **B. Rachunok**, R. Flage, and R. Nateghi. "Mapping climate discourse to climate opinion: An approach for augmenting surveys with social media to enhance understandings of climate opinion in the United States". In: *PLOS ONE* (2021). DOI: 10.1371/journal.pone.0245319.

[18] M. Choi, **B. Rachunok**, and R. Nateghi. "Short-term solar irradiance forecasting using convolutional neural networks and cloud imagery". In: *Environmental Research Letters* 16.4 (2021). DOI: 10.1088/1748-9326/abe06d.

[17] R. Obringer, **B. Rachunok**, D. Maia-Silva, M. Arbabzadeh, R. Nateghi, and K. Madani. "The overlooked environmental footprint of increasing Internet use". In: *Resources, Conservation and Recycling* 167 (2021). DOI: 10.1016/j.resconrec.2020.105389.

[16] **B. Rachunok**, J. Bennett, R. Flage, and R. Nateghi. "A path forward for leveraging social media to improve the study of community resilience". In: *International Journal of Disaster Risk Reduction* 59 (2021). DOI: 10.1016/j.ijdrr.2021.102236.

[15] **B. Rachunok** and R. Nateghi. "Overemphasis on recovery inhibits community transformation and creates resilience traps". In: *Nature Communications* 12.1 (2021). DOI: 10.1038/s41467-021-27359-5.

[14] N. Alemazkoor, **B. Rachunok**, D. R. Chavas, A. Staid, A. Louhghalam, R. Nateghi, and M. Tootkaboni. "Hurricane-induced power outage risk under climate change is primarily driven by the uncertainty in projections of future hurricane frequency". In: *Scientific reports* 10.1 (2020). DOI: 0.1038/s41598-020-72207-z.

[13] R. Kumar *, **B. Rachunok** *, D. Maia-Silva, and R. Nateghi. "Asymmetrical response of California electricity demand to summer-time temperature variation". In: *Scientific Reports* (2020). DOI: 10.1038/s41598-020-67695-y.

[12] A. Paulvannan Kanmani, R. Obringer, **B. Rachunok**, and R. Nateghi. "Assessing Global Environmental Sustainability Via an Unsupervised Clustering Framework". In: *Sustainability* (2020). DOI: 10.3390/su12020563.

[11] **B. Rachunok** and R. Nateghi. "The sensitivity of electric power infrastructure resilience to the spatial distribution of disaster impacts". In: *Reliability Engineering & System Safety* (2020). DOI: 10.1016/j.ress.2019.106658.

[10] **B. Rachunok**, A. Staid, J.-P. Watson, and D. L. Woodruff. "Assessment of wind power scenario creation methods for stochastic power systems operations". In: *Applied Energy* 268 (2020). DOI: 10.1016/j.apenergy.2020.114986.

[9] **B. Rachunok**, J. Bennett, and R. Nateghi. "Twitter and Disasters: A Social Resilience Fingerprint". In: *IEEE Access* (2019). DOI: 10.1109/ACCESS.2019.2914797.

Manuscripts Under Review & Preprints

* indicates equal contributions, Underline indicates an advisee

[8] J. Chala Lancheros and **B. Rachunok**. "Prepare vs Response: Stochastic Modeling of Flood Resilience in NC Rural Schools". In: *Risk Analysis* (2026).

[7] Y. Essus, R. R. Vatsavai, and **B. Rachunok**. "Data Leakage Inflates Generalizability of Power Outage Prediction Models". In: *arXiv preprint* (2026). DOI: 10.48550/arXiv.2608.24665. arXiv: 2608.24665.

[6] T. Groton and **B. Rachunok**. "Population-level measures of perceived food access reveal barriers beyond geographic proximity". In: *Nature Food* (2026). DOI: 10.48550/arXiv.2609.12132. arXiv: 2609.12132.

[5] E. Harris and **B. Rachunok**. "Clustering-based Assessment of Power System and Energy System Inderdependence". In: *Reliability Engineering and System Safety* (2026).

[4] R. Smith, B. McConnell, R. King, and **B. Rachunok**. "Two for One: Multi-Objective Disaster Recovery Models Improve Public Utility Post Disaster". In: *International Journal of Disaster Risk Reduction* (2026).

[3] R. Smith, **B. Rachunok**, B. McConnell, and R. King. "Fast is Fair: Trade-offs in Cell Tower Network Restoration". In: *Reliability Engineering and System Safety* (2026).

[2] A. Biswas, A. Helmrich, and **B. Rachunok**. "Flood-related Tweets offer High-resolution impact assessments". In: *npj Urban Sustainability* (2025).

[1] I. Sepulveda and **B. Rachunok**. "Comparing online and offline Social Capital". In: *International Journal of Disaster Risk Reduction* (2025).

Selected Honors & Awards

KIETS Climate Leaders Program
Kenan Institute for Engineering, Technology & Science *2026*
Selected with Ph.D. student Juan Chala as a 2026 Student/Mentor team.

Best Paper Award, Foundations of Risk
Society for Risk Analysis *2025*
Award given to Ph.D. Student Yamil Essus's paper on ecological resilience.

C.A. Anderson Outstanding Faculty Award
NC State ISE Department *2024*
Award given to one faculty per year as voted by graduating seniors

Sustainable Society Conference Delegate
University Global Partnership Network *2024*
One of 8 faculty selected from NC State to attend

Rising Environmental Leadership Program Fellow
Stanford Woods Institute for the Environment *2021*

Outstanding Research Award
Purdue University School of Industrial Engineering *2020*

Outstanding Service Award
Purdue University School of Industrial Engineering *2020*

Student Spotlight
Society for Risk Analysis Engineering and Infrastructure Specialty Group *2020*

Lee Chaden Fellowship
Purdue University School of Industrial Engineering *2018*

Estus and Vashti Magoon Award
Purdue University *2018*

Research Advising

Current PhD Student Advisees

Ignacio Sepulveda — **Spring 2023-present**
NC State University — *Industrial & Systems Engineering*

Shakhawath Hossain — **Fall 2025-present**
NC State University — *Industrial & Systems Engineering*

Camryn Deames — **Fall 2025-present**
NC State University — *Business Management*
Co-advised with Dr. Brandon McConnell

Elizabeth Harris — **Fall 2025-present**
NC State University — *Operations Research*

Caroline Folsom — **Fall 2025-present**
NC State University — *Operations Research*

Juan Chala — **Fall 2025-present**
NC State University — *Industrial & Systems Engineering*

Current Master's Student Advisees

Zackaria Radwan — **Fall 2026-present**
NC State University — *Operations Research*

PhD Alumni

Yamil Essus — **Spring 2023-Spring 2026**
NC State University — *Industrial & Systems Engineering*
Current Position: Postdoctoral Fellow, University of Toronto

Robert Smith — **Spring 2024-Spring 2026**
NC State University — *Operations Research*
Co-chaired with Dr. Brandon McConnell & Dr. Russell King. Current Position: Postdoctoral Research Associate, Institute for Risk Management and Insurance Innovation, UNC–Chapel Hill

Austin Semmel — **Spring 2024-Spring 2026**
NC State University — *Operations Research*
Co-chaired with Dr. Brandon McConnell & Dr. Sebastian Heese. Current Position: U.S. Military Academy at West Point

Master's Alumni

Teresa Groton — **Fall 2024-Spring 2026**
NC State University — *Operations Research*
Current Position: U.S. Military Academy at West Point

Patrick Leavitt — **Fall 2023-Spring 2025**
NC State University — *Industrial & Systems Engineering*
Current Position: Data Scientist, U.S. Coast Guard

NC State Graduate Advisory Committees

Edric Pham — **Fall 2026-present**
NC State University, Committee Member — *Graduate Advisory Committee*

Madison Horgan — **Fall 2026-present**
NC State University, Committee Member — *Graduate Advisory Committee*

Ryan Andersen — **Summer 2026-present**
NC State University, Committee Member — *Graduate Advisory Committee*

Verónica Díaz-Pacheco — **Summer 2026-present**
NC State University, Ph.D. Committee Member — *Operations Research*

Cameron Lisy — **Summer 2026-present**
NC State University, Ph.D. Committee Member — *Operations Research*

Laura Mora — **Spring 2026-present**
NC State University, Committee Member — *Graduate Advisory Committee*

Roya Sahraei — **Spring 2026-present**
NC State University, Committee Member — *Graduate Advisory Committee*

Lindsey Jacobson — **Fall 2025-present**
NC State University, Committee Member — *Graduate Advisory Committee*

Yongseok Jeon — **Spring 2025-present**
NC State University, Committee Member — *Graduate Advisory Committee*

Lucia Rebeca Gomez Hurtado — **Spring 2025-present**
NC State University, Committee Member — *Graduate Advisory Committee*

Archit Sunil Kulkarni — **Spring 2025-present**
NC State University, Committee Member — *Graduate Advisory Committee*

Troy Wibowo — **Spring 2024-present**
NC State University, Ph.D. Committee Member — *Industrial & Systems Engineering*

Graduate School Representative Service

Clayson Shumway — **Summer 2025-present**
NC State University, Graduate School Representative — *Economics & Management*

External Graduate Advisory Committees

Cameron Moore — **Spring 2026-present**
Duke University, Ph.D. Committee Member — *Graduate Advisory Committee*
Advisor: Dr. Marta Zaniolo

Swagato Biswas Ankon — **Spring 2023-present**
University of Georgia, MS Committee Member — *Environmental and Water Resources Engineering*

Completed Graduate Advisory Committees

Pranav Jain — **Spring 2023-Spring 2024**
NC State University, Ph.D. Committee Member — *Industrial & Systems Engineering*

Graduate Independent Study Advisees

Aakash Dhruv — **Spring 2024**
NC State University — *Integrated Manufacturing Systems Engineering Institute*
Current Position: Supply Chain Engineering, Amazon

Current Undergraduate Research Advisees

Suyash Sapre — **2026-present**
NC State University — *Industrial & Systems Engineering*

Greta Perez — **2026-present**
NC State University — *Industrial & Systems Engineering*

Benjamin Lake — **2026-present**
NC State University — *Industrial & Systems Engineering*

Previous Undergraduate Research Advisees

Audrey Westlund — **Spring 2025-2026**
NC State University — *Industrial & Systems Engineering*

Ashton Sumpter — **Spring 2025-2026**
NC State University — *Industrial & Systems Engineering*

Matthew Bogdan — **Fall 2024-2026**
NC State University — *Industrial & Systems Engineering*

James Wilt — **Summer 2024-2026**
NC State University — *Industrial & Systems Engineering REU*

Ashelynn Jacobson — **Summer 2024-2026**
NC State University — *Industrial & Systems Engineering REU*

Drew Karnbad — **Spring 2024-2026**
NC State University — *Industrial & Systems Engineering*

Mike Mueller — **Spring 2024**
NC State University — *Industrial & Systems Engineering*

Victoria Mendez Gonzalez — **Summer 2023**
University of Puerto Rico at Mayagüez — *Industrial & Systems Engineering*

Fatemeh Heydari — **Summer 2023**
NC State University — *MI-REU Program*

Estevan Zamora — **Summer 2023**
NC State University — *MI-REU Program*

Julia Seidel — **Spring 2023**
NC State University — *Honors Thesis*

Adam Nayak — **2021-2022**
Stanford University

Policy Engagement & Government Briefings

Fayetteville City Council Work Session — **Fayetteville, NC**
North Carolina Data Center Water Resource Analysis — *August 2026*
Invited policy briefing on data-center water, energy, wastewater, and infrastructure impacts

North Carolina Energy Policy Task Force — **Raleigh, NC**
Initial Assessment Plan: North Carolina Data Center Water Resource Analysis — *July 2026*

Teaching

ISE408: Design and Control of Production and Service Systems — *Spring 2023-Fall 2026*

ISE589: Computational Applications of Sustainability and Resilience — *Fall 2023, Fall 2024, Fall 2025, Fall 2026*

Invited Seminars

ECo Talks, The Earth Commons — **Georgetown University**
The Myth of Access: Affordability, Perception, and the Reality of Infrastructure Equity — *April 2026*

CEE Graduate Colloquium — **Duke University**
The Myth of Access: Quantifying Access to Essential Services Using AI, Durham, NC — *February 2026*

Civil and Environmental Engineering — **Stanford University**
Risk, Optimization, and AI: Quantifying Access to Essential Services — *February 2026*

Environmental, Water, and Coastal Engineering Seminar Series — **NC State University**
Quantifying Non-Spatial Dimensions of Access to Essential Services, Raleigh, NC — *December 2025*

Society for Risk Analysis Risk Science Webinar Series — **Virtual**
Quantifying and Operationalizing Resilience, Equality and Equity in Power Systems — *April 2025*

AI for Extreme Weather Event and Disaster Risk Management Seminar Series — **Virtual**
Quantifying Community Resilience — *October 2024*

Center for Environmental and Resource Economic Policy — **NC State University**
Engineering vs. Ecological Resilience — *April 2024*

International Risk and Resilience Research Alliance — **Virtual**
The Pitfalls of Aggregation in Assessing Community Resilience — *February 2024*

Energy Data Analytics Symposium — **Duke University**
Ethics & Social Responsibility of Data Analytics — *April 2023*

Operations Research Seminar Series — **NC State University**
Linking Sustainability & Resilience with Operations Research — *March 2023*

Civil & Environmental Engineering — **Cornell University**
Avoiding Resilience Traps with Unsupervised Learning — *March 2022*

Industrial & Systems Engineering — **NC State University**
Methods for Operationalizing Community Resilience Using Big Data — *January 2022*

INFORMS Student Chapter Summer Research Series — **Oklahoma University**
Social Resilience Fingerprinting — *April 2020*

Invited Conference Presentations

Society for Risk Analysis Europe Annual Conference — **Alicante, Spain**
Beyond Geography: Population-Scale Measurement of A-Spatial Food Access Using Large Language Models — *May 2026*

Society for Risk Analysis Annual Meeting — **Washington, DC**
Incorporating Equity and Fairness into a Mathematical Model for Power Restoration After Natural Hazards — *December 2025*

Society for Risk Analysis Annual Meeting — **Washington, DC**
The Future of Risk Analysis: Perspectives from Emerging Risk Professionals — *December 2025*

INFORMS Annual Meeting — **Atlanta, GA**
Incorporating Equity and Fairness into a Mathematical Model for Power Restoration After Natural Hazards — *October 2025*

Society for Risk Analysis Europe Annual Conference — **Stavanger, Norway**
Post-disaster Power System Restoration Optimization — *June 2025*

Society for Risk Analysis Annual Meeting — **Austin, TX**
Integrating Theories of Community Resilience — *December 2024*

INFORMS Annual Meeting **Phoenix, AZ**
Predicting and Understanding Residential Water use with Interpretable Machine Learning *October 2023*

The International Symposium on Sustainable Systems and Technology **Fort Collins, CO**
Quantifying Community Resilience to Break Resilience Traps *June 2023*

Institute for Industrial and Systems Engineers Annual Conference **New Orleans, LA**
Quantifying Community Transformation *May 2023*

AGU Fall Meeting **New Orleans, LA**
Socio-hydrological impacts of water scarcity on household water affordability *December 2022*

Society for Risk Analysis Annual Meeting **Tampa, FL**
Socio-hydrological impacts of water scarcity on household water affordability *December 2022*

Society for Risk Analysis Annual Meeting **Virtual**
Quantifying Community Resilience to Break Resilience Traps *December 2020*

International Conference on Computational Social Science (IC2S2). **Virtual**
Be Careful Where You Get Your Tweets: Bias in Analysis from Tweet Acquisition Source *June 2020*

INFORMS Annual Meeting **Virtual**
Quantifying Community Transformation *October 2020*

Natural Hazards Center Researcher's Meeting **Virtual**
Natural Hazard Twitter Analysis Depends on Tweet Sources *May 2020*

INFORMS Annual Meeting **Seattle, WA**
Social Resilience Fingerprinting *October 2019*

Institute for Industrial and Systems Engineers Annual Conference **Orlando, FL**
Interdependent Infrastructure System Risk & Resilience to Natural Hazards. *May 2019*

Society for Risk Analysis Annual Meeting **New Orleans, LA**
Electric Power Infrastructure Resilience *December 2018*

INFORMS Annual Meeting **Phoenix, AZ**
The Sensitivity of Electric Power Infrastructure Resilience to the Spatial Distribution of Disaster Impacts *October 2018*

IEEE International Conference on Probabilistic Methods Applied to Power 2018 Systems **Boise, ID**
Stochastic Unit Commitment Performance Considering Monte Carlo Wind Power Scenarios *June 2018*

Society for Risk Analysis Annual Meeting **Crystal City, VA**
Transportation Network Recovery Analysis, an Equilibrated Perspective *December 2017*

INFORMS Annual Meeting **Houston, TX**
The Cost of Wrong Assumptions in Tropical Cyclone Frequency Distributions *October 2017*

Institute for Industrial and Systems Engineers Annual Conference **Anaheim, CA**
UAVs Provide Lifesaving Medical Care *May 2016*

Student Conference Presentations

Underline indicates an advisee

Society for Risk Analysis Annual Meeting **Washington, DC**
Characterization of the Dependency Between Power and Communication Outages *December 2025*
E. Harris and **B. Rachunok**

Society for Risk Analysis Annual Meeting **Washington, DC**
Expanding the Dimensions of Food Access: Linking Experience and Geography Through Data-Driven Analysis *December 2025*
T. Groton and **B. Rachunok**

93rd Military Operations Research Society Symposium **Leesburg, VA**
A Framework for Analyzing Operational Efficiency in US Army Aviation *June 2025*
Self-Organizing Map-based Clustering of Flight Dispatch Decisions
A. Semmel, H. S. Heese, B. McConnell, M. Powell, and **B. Rachunok**
Selected Best in Working Group (Readiness)

Society for Risk Analysis Europe Annual Conference **Stavanger, Norway**
Quantifying Ecological Resilience *June 2025*
Y. Essus and **B. Rachunok**

Society for Risk Analysis Annual Meeting **Austin, TX**
Fairness in Post-Disaster Power Restoration *December 2024*
I. Sepulveda and **B. Rachunok**

Society for Risk Analysis Annual Meeting **Austin, TX**
Quantifying Ecological Resilience *December 2024*
Y. Essus and **B. Rachunok**

INFORMS Annual Meeting **Seattle, WA**
Fairness in Post-Disaster Power Restoration *October 2024*
I. Sepulveda and **B. Rachunok**

INFORMS Annual Meeting **Seattle, WA**
Estimating Access risk during blackouts due to vehicle electrification in North Carolina *October 2024*
Y. Essus and **B. Rachunok**

International Symposium on Sustainable Systems and Technology **Baltimore, MD**
Quantifying Social Capital *June 2024*
I. Sepulveda and **B. Rachunok**

International Symposium on Sustainable Systems and Technology **Baltimore, MD**
Electric Vehicles Limit Equitable Access to Essential Services During Blackouts *June 2024*
Y. Essus and **B. Rachunok**

Society for Risk Analysis Annual Meeting **Washington, DC**
Quantifying Social Capital *December 2023*
I. Sepulveda and **B. Rachunok**

Society for Risk Analysis Annual Meeting **Washington, DC**
Electric Vehicles Limit Equitable Access to Essential Services During Blackouts *December 2023*
Y. Essus and **B. Rachunok**

INFORMS Annual Meeting **Phoenix, AZ**
Electric Vehicles Limit Equitable Access to Essential Services During Blackouts *October 2023*
Y. Essus and **B. Rachunok**

Scientific Community Service

Reviewer

Journal Articles

Communications Engineering, Journal of Infrastructure Systems, Journal of Water Resources Planning and Management, Nature Communications, IEEE Access, Risk Analysis, International Journal of Disaster Risk Reduction, Reliability Engineering and System Safety, Utility Policy, Journal of Management in Engineering

External Grant Proposals

DOE Science Synthesis Prize, November 2024
NSF Human-Environment Geographical Sciences Program, ad hoc, May 2023
NSF Human-Environment Geographical Sciences Program, ad hoc, January 2023

Conference Organizing

Planning Committee Member
International Symposium on Sustainable Systems and Technology (ISSST) *2026*

Sustainability and Resilience of Infrastructure Track Chair
International Symposium on Sustainable Systems and Technology (ISSST) *2024*

Equity, Sustainability and Engineering Session Chair
Society for Risk Analysis Annual Meeting *2023*

Professional Societies

Society for Risk Analysis *2016-present*
Council member 2021-2023; Secretary 2024-present; Board Liaison, Risk Science & Translation Committee, 2026-present

INFORMS *2017-present*

Institute for Industrial and Systems Engineers *2016-present*

University Service

Department and University Committees

ISE Masters of Engineering Management Committee *Fall 2023, Spring 2024*

ISE PhD Admissions Committee *Spring 2023, Fall 2024, Spring 2025*

Other Service

Engineering Open House *Spring 2023, Fall 2023, Spring 2024, Fall 2024*

ISE Summer Camp *Summer 2023, Summer 2024*